# MTH2010

## UNIVERSITY OF EXETER

## FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

## MATHEMATICS AND STATISTICS

### January 2023

**Groups, Rings and Fields**

**Module Leader: Mohamed Saïdi**

**Duration: 2 HOURS**

**Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).**

*Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.*

This is a **CLOSED NOTE** examination.

## SECTION A

**All rings are commutative with an identity element** $1$ **and** $1 \neq 0$**.**

1. (a) Let $\phi : G \to H$ be a group homomorphism and $K$ a cyclic subgroup of $G$. Show that

$$\phi(K) = \{h \in H : h = \phi(g),\ g \in K\}$$

is a cyclic subgroup of $H$. (10)

(b) Let $D_{12}$ be the dihedral group with 12 elements. Is there a surjective homomorphism from $D_{12}$ to a group with 4 elements? Justify your answer and construct such a homomorphism if it exists. (10)

(c) Let $R$ be a ring.

(i) Define what it means for a nonempty subset $I$ of $R$ to be an ideal, and a prime ideal. (6)

(ii) Let $I$ and $J$ be ideals of $R$. Define

$$I + J = \{a + b : a \in I,\ b \in J\}.$$

Show that $I + J$ is an ideal of $R$ and $I + J$ is the smallest ideal of $R$ containing both $I$ and $J$. (10)

(iii) Assume $R = \mathbf{Z}$ is the ring of integers. Let $I$ and $J$ be nonzero distinct prime ideals of $\mathbf{Z}$. Show that $I + J = \mathbf{Z}$. (6)

(d) Define what a unit of a ring $R$ is. Show that the units of the ring of polynomials $\mathbf{Q}[X]$ are the nonzero constant polynomials $a \in \mathbf{Q}$, $a \neq 0$. Here $\mathbf{Q}$ is the field of rational numbers (8)

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## SECTION B

**All rings are commutative with an identity element** $1$ **and** $1 \neq 0$**.**

2. Let $(\mathbf{Q}, +)$ be the group of rational numbers with the addition operation.

   (a) Show that the subset $\mathbf{Z}$ of $\mathbf{Q}$ consisting of integers is a subgroup of $(\mathbf{Q}, +)$. (4)

   (b) For $x \in \mathbf{Q}$ let
   $$x + \mathbf{Z} = \{x + n : n \in \mathbf{Z}\}$$
   be the left coset $x + \mathbf{Z}$ of $\mathbf{Z}$ in $\mathbf{Q}$.

   Show that $x + \mathbf{Z} = a + \mathbf{Z}$ for a unique rational number $a$ with $0 \leq a < 1$. (7)

   (c) Define what the quotient group $\mathbf{Q}/\mathbf{Z}$ and the group operation of $\mathbf{Q}/\mathbf{Z}$ are. Explain what the inverse of an element of $\mathbf{Q}/\mathbf{Z}$ is. Show that $\mathbf{Q}/\mathbf{Z}$ is infinite. (7)

   (d) Show that every element of the factor group $\mathbf{Q}/\mathbf{Z}$ has finite order, and $\mathbf{Q}/\mathbf{Z}$ contains elements of order $k$ for any positive integer $k$. (7)

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3. (a) Let $p$ and $q$ be distinct prime numbers and $G$ a group of cardinality $p^2q$. Show that $G$ has either a normal $p$-Sylow subgroup or a normal $q$-Sylow subgroup.

   You can use without proof the fact that a finite group of cardinality 12 which doesn't have a normal 3-Sylow subgroup is isomorphic to the alternating group $A_4$. (8)

   (b) Let $R$ be a ring. We say an element $r \in R$ is nilpotent if $r^m = 0$ for some integer $m \geq 1$. Let $r \in R$ be a nilpotent element.

   (i) Prove that $r$ is either 0 or a zero divisor of $R$. (4)

   (ii) Prove that $rs$ is nilpotent for every element $s$ of $R$. (4)

   (iii) Prove that $1 + r$ is a unit of $R$. (Hint: find an inverse of $1 + r$ in case $r^2 = 0$, $r^3 = 0, \ldots$, then find a pattern.) (5)

   (iv) Let $u$ be a unit of $R$. Show that $r + u$ is a unit of $R$. (4)

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4. (a) Let $\mathbf{Z}$ be the ring of integer numbers and $R$ the ring

$$\mathbf{Z}[\sqrt{7}] = \{a + b\sqrt{7} : a, b \in \mathbf{Z}\}$$

and

$$I = (3, 1 + \sqrt{7})_R$$

the ideal of $R$ generated by 3 and $1 + \sqrt{7}$.

By constructing a surjective ring homomorphism from $R$ to a field $F$ with kernel $I$, or otherwise, show that $I$ is a maximal ideal of $R$. Clearly state any theorems from the lecture you use. (13)

(b) Let $G$ be a finite group of cardinality $|G| = m$ an odd integer. Let $x$ be an element of $G$ distinct from the neutral element $e_G$. Show that the elements $x$ and $x^{-1}$ cannot be conjugate in $G$. (12)

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